QUESTION ANALYSIS

EEE421

Solid State Devices

Question Analysis

Sessions 2020–21 *1 papers, 24 questions, 45 formulas*

INSIDE

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 24 questions and 145 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Semiconductor in equilibrium	30	21 %	5	Ionization (15); Fermi level (5); Compensation (5)
Carrier transport	25	17 %	4	Mobility (10); Hall effect (10); Einstein relation (5)
PN junction	25	17 %	4	Depletion width (10); Biased junction (10); Band diagram (5)
Excess carriers	20	14 %	3	Steady state (10); Minority carriers (5); Carrier pulse (5)
Quantum basics	15	10 %	2	Fermi probability (10); Density of states (5)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	All
Quantum basics	**15**	**15**
Fermi probability	**10**	10
Density of states	**5**	5
Semiconductor in equilibrium	**30**	**30**
Ionization	**15**	15
Fermi level	**5**	5
Compensation	**5**	5
Carrier concentrations	**5**	5
Carrier transport	**25**	**25**
Mobility	**10**	10
Hall effect	**10**	10
Einstein relation	**5**	5
Excess carriers	**20**	**20**
Steady state	**10**	10
Minority carriers	**5**	5
Carrier pulse	**5**	5
PN junction	**25**	**25**
Depletion width	**10**	10
Biased junction	**10**	10
Band diagram	**5**	5
Tunnel diode	**5**	**5**
Operation	**5**	5
MOS structure	**15**	**15**
Band diagrams	**5**	5
Double gate	**5**	5
Inversion	**5**	5
Bipolar transistor	**10**	**10**

Topic and subtopic	2020-21	All
Operating modes	**5**	5
Minority distribution	**5**	5

3. Priority list:

Must know: Semiconductor in equilibrium (30), Carrier transport (25), PN junction (25), Excess carriers (20), Quantum basics (15), MOS structure (15), Bipolar transistor (10), Tunnel diode (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Quantum basics:

15 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 3.

Most often numerical (2 of 2). Subtopics: Density of states; Fermi probability.

Formulas:

Subtopic	Formula	Equation	Used in
Density of states (5)	**Density of states**	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$	20-21 1(c)
	States from E_c to $E_c + kT$	$N = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$	20-21 1(c)
Fermi probability (10)	Energy for a given probability	$E - E_F = kT \ln\left(\frac{1}{f_F} - 1\right)$	20-21 2(d)

Symbols:

g_c = states per unit energy per unit volume ↩

Questions:

No.	Paper	Marks	Type	Question
Density of states				
1	20-21 1(c)	5	numerical	Determine the number of quantum states(cm^{-3}) in silicon between E_c and E_c+kT at $T = 300$ K.
Fermi probability				
2	20-21 2(d)	10	numerical	Answer the following questions on Fermi level: (i) At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10% ? (ii) Find the electron concentration(n_0) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ cm^{-3}).

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4.2. Semiconductor in equilibrium:

30 marks, 5 questions, in 1 of 1 papers. Book: Neamen Ch. 4.

Most often theory (2 of 5). Subtopics: Fermi level; Ionization; Compensation; Carrier concentrations.

Formulas:

Subtopic	Formula	Equation	Used in
Fermi level (5)	**Fermi–Dirac function**	$f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$	20-21 1(a)
	Electron concentration	$n_0 = N_c \exp\left[-\dfrac{E_c-E_F}{kT}\right]$	20-21 1(a)
	Hole concentration	$p_0 = N_v \exp\left[-\dfrac{E_F-E_v}{kT}\right]$	20-21 1(a)
	Mass-action law	$n_0 p_0 = {n_i}^2 = N_c N_v \exp\left(-\dfrac{E_g}{kT}\right)$	20-21 1(a)
	Fermi level vs doping	$E_F - E_{Fi} = kT\ln\dfrac{N_d}{n_i}$ (n-type), $E_{Fi} - E_F = kT\ln\dfrac{N_a}{n_i}$ (p-type)	20-21 1(a)
Ionization (15)	N_c at another temperature	$N_c(T) = N_c(300)\left(\dfrac{T}{300}\right)^{3/2}$	20-21 1(d)
	Fermi level from E_c	$E_c - E_F = kT\ln\dfrac{N_c}{n_0}$	20-21 1(d)
	Donor occupation	$f_d = \dfrac{1}{1+\frac{1}{2}\exp\left(\dfrac{E_d-E_F}{kT}\right)}$	20-21 1(d)
	Fraction not ionized	$\dfrac{n_d}{n_d+n_0} = \dfrac{1}{1+\dfrac{N_c}{2N_d}\exp\left[-\dfrac{E_c-E_d}{kT}\right]}$	20-21 1(d), 20-21 6(b)
	Freeze-out estimate	$n_0 \approx \sqrt{\dfrac{N_c N_d}{2}}\exp\left[-\dfrac{E_c-E_d}{2kT}\right]$	20-21 1(d)
Carrier concentrations (5)	**Charge neutrality**	$n_0 + N_a^- = p_0 + N_d^+$	20-21 3(a)
	Electrons, compensated	$n_0 = \dfrac{N_d-N_a}{2} + \sqrt{\left(\dfrac{N_d-N_a}{2}\right)^2 + n_i^2}$	20-21 3(a)
	Holes, compensated	$p_0 = \dfrac{N_a-N_d}{2} + \sqrt{\left(\dfrac{N_a-N_d}{2}\right)^2 + n_i^2}$	20-21 3(a)

Symbols:

f_F = probability that a state at energy E holds an electron ↩

E_F = Fermi level ↩

N_c = effective density of states in the conduction band ↩

N_v = effective density of states in the valence band ↩

n_i = intrinsic carrier concentration ↩

E_{Fi} = intrinsic Fermi level (near midgap) ↩

E_d = donor level ↩

n_d = electrons still in donor states ↩

N_a^-, N_d^+ = ionized acceptors and donors (cm^{-3}) ↩

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Questions:

No.	Paper	Marks	Type	Question
Fermi level				
1	20-21 1(a)	5	theory	Sketch the position of Fermi level in Si at 300 K as a function of doping concentration.
Ionization				
2	20-21 1(d)	10	numerical	A Si sample is doped with $N_d = 1 \times 10^{15}$ cm^{-3} of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. E_c/E_v) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner *[sic: should be donor]* states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ cm^{-3}, for Si].
3	20-21 6(b)	5	numerical	Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer.
Compensation				
4	20-21 2(a)	5	theory	Draw energy band diagram of a compensated semiconductor.
Carrier concentrations				
5	20-21 3(a)	5	derivation	Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for n_0 and p_0 in terms of N_a, N_d, and n_i.

4.3. Carrier transport:

25 marks, 4 questions, in 1 of 1 papers. Book: Neamen Ch. 5.

Most often theory (3 of 4). Subtopics: Einstein relation; Mobility; Hall effect.

Formulas:

Subtopic	Formula	Equation	Used in
Einstein relation (5)	**Einstein relation**	$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$	20-21 2(b)
Mobility (10)	**Drift velocity**	$v_d = \mu E$	20-21 3(c)
	Mobility	$\mu = \frac{e\tau}{m^*}$	20-21 3(c)
	Conductivity	$\sigma = e(n\mu_n + p\mu_p)$	20-21 3(c)
	Temperature and mobility	$\mu_L \propto T^{-3/2}$, $\mu_I \propto \frac{T^{3/2}}{N_I}$, $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$	20-21 4(a)
Hall effect (10)	**Hall voltage**	$V_H = \frac{I_x B_z}{epd}$ (p-type), $V_H = -\frac{I_x B_z}{end}$ (n-type)	20-21 5(d)
	Carrier density, Hall	$n = -\frac{I_x B_z}{e d V_H}$	20-21 5(d)
	Hall mobility	$\mu_n = \frac{I_x L}{e n V_x W d}$	20-21 5(d)

Symbols:

μ = mobility (cm^2/V s) ↩

E = electric field (V/cm) ↩

τ = mean time between collisions (s) ↩

m^* = conductivity effective mass ↩

μ_L, μ_I = lattice- and impurity-limited mobility ↩

N_I = ionized impurity concentration ↩

d = thickness along B_z

L = length along x, W = width along y

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Questions:

No.	Paper	Marks	Type	Question
Einstein relation				
1	20-21 2(b)	5	derivation	Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation.
Mobility				
2	20-21 3(c)	5	theory	Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of τ.
3	20-21 4(a)	5	theory	Explain the effects of temperature on carrier mobility inside a semiconductor.
Hall effect				
4	20-21 5(d)	10	theory	Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. (i) What is the Hall voltage? (ii) What is type of carrier? (iii) What is the carrier concentration? (iv) What is the mobility?

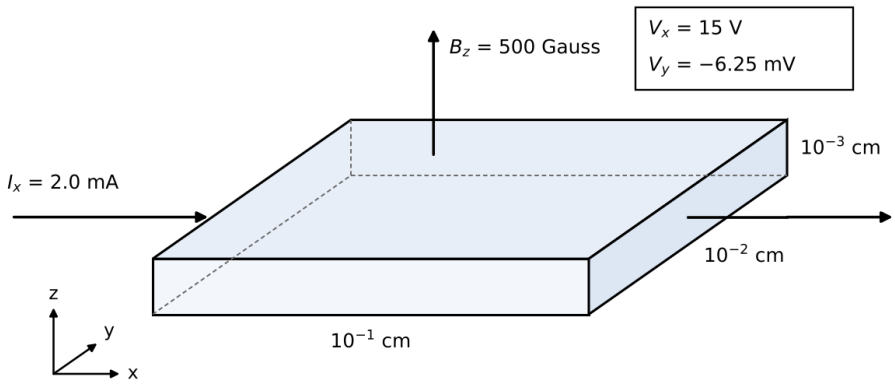


Figure 1: 20-21 5(d): Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

4.4. Excess carriers:

20 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 6.

Most often theory (2 of 3). Subtopics: Minority carriers; Steady state; Carrier pulse.

Formulas:

Subtopic	Formula	Equation	Used in
Minority carriers (5)	**Diffusion current**	$J_n = eD_n\frac{dn}{dx},\ J_p = -eD_p\frac{dp}{dx}$	20-21 1(b)
	Poisson's equation	$\frac{d^2\phi}{dx^2} = -\frac{\rho(x)}{\epsilon_s} = -\frac{dE}{dx}$	20-21 1(b)
Steady state (10)	**Ambipolar transport**	$D'\frac{\partial^2(\delta n)}{\partial x^2} + \mu' E\frac{\partial(\delta n)}{\partial x} + g' - R' = \frac{\partial(\delta n)}{\partial t}$	20-21 3(d)
	Steady-state diffusion	$\delta n(x) = \delta n(0)\exp\left(-\frac{x}{L_n}\right),\ L_n = \sqrt{D_n\tau_{n0}}$	20-21 3(d)
Carrier pulse (5)	n-type, low injection	$D_p\frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0\frac{\partial(\delta p)}{\partial x} - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$	20-21 5(a)
	Moving pulse	$\delta p(x,t) = \frac{N_0}{\sqrt{4\pi D_p t}}\exp\left(-\frac{t}{\tau_{p0}}\right)\exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$	20-21 5(a)

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Symbols:

D_n, D_p = diffusion coefficients (cm^2/s) ↩

ϕ = potential (V) ↩

ρ = charge density (C/cm^3) ↩

L_n = minority-electron diffusion length ↩

N_0 = pairs created per unit area at $x = 0$, $t = 0$ ↩

Questions:

No.	Paper	Marks	Type	Question
Minority carriers				
1	20-21 1(b)	5	theory	How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction?
Steady state				
2	20-21 3(d)	10	numerical	For the following questions: (i) Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ cm^{-3} for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ cm^{-3}). (ii) Find the distance where the excess charge will drop to 10×10^{13} cm^{-3}. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ cm^2 s^{-1}, $D_p = 25$ cm^2 s^{-1}) (iii) Plot n and p as function of x in log scale.
Carrier pulse				
3	20-21 5(a)	5	theory	Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field E_0. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time.

4.5. PN junction:

25 marks, 4 questions, in 1 of 1 papers. Book: Neamen Ch. 7, 8.

Most often numerical (2 of 4). Subtopics: Depletion width; Band diagram; Biased junction.

Formulas:

Subtopic	Formula	Equation	Used in
Depletion width (10)	Charge balance	$N_a x_p = N_d x_n$	20-21 2(c)
	Depletion width	$W = x_n + x_p = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2}$	20-21 2(c)
	Each side	$x_n = W \frac{N_a}{N_a + N_d}$, $x_p = W \frac{N_d}{N_a + N_d}$	20-21 2(c)
	Peak field	$E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s} = \frac{2V_{bi}}{W}$	20-21 6(d)
Band diagram (5)	**Fermi potentials**	$\phi_{Fp} = V_t \ln \frac{N_a}{n_i}$, $\lvert\phi_{Fn}\rvert = V_t \ln \frac{N_d}{n_i}$	20-21 3(b)
	Built-in potential	$V_{bi} = \lvert\phi_{Fn}\rvert + \lvert\phi_{Fp}\rvert = V_t \ln \frac{N_a N_d}{n_i^2}$	20-21 3(b), 20-21 6(d)
Biased junction (10)	**Law of the junction**	$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$, $n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$	20-21 4(d)
	Majority carriers at the edge	$n_n = N_d + \delta p_n$	20-21 4(d)

Symbols:

x_n, x_p = depletion widths on the n and p sides ↩

Questions:

No.	Paper	Marks	Type	Question
Depletion width				
1	20-21 2(c)	0	derivation	Using the expression for V_{bi}, find the width of the depletion region.
2	20-21 6(d)	10	numerical	A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{15}$ cm^{-3}. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ *[sic: should be* $\epsilon_r = 11.7$*]*)
Band diagram				
3	20-21 3(b)	5	theory	Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, E_C, E_V, E_F, E_{Fi}, $e\phi_{Fn}$, $e\phi_{Fp}$ and eV_{bi}.
Biased junction				
4	20-21 4(d)	10	numerical	Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm^{-3}, assume doping in N side is 1×10^6 cm^{-3} *[sic: should be* 1×10^{16} *cm*$^{-3}$*]*. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side.

4.6. Tunnel diode:

5 marks, 1 question, in 1 of 1 papers. Book: Neamen Ch. 8, 15.

Most often theory (1 of 1). Subtopics: Operation.

Questions:

No.	Paper	Marks	Type	Question
Operation				
1	20-21 5(c)	5	theory	Explain the working principle of Tunnel diode. Draw its characteristics curve.

4.7. MOS structure:

15 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 10.

Most often theory (3 of 3). Subtopics: Band diagrams; Double gate; Inversion.

Formulas:

Subtopic	Formula	Equation	Used in
Band diagrams (5)	**Fermi potential, p substrate**	$\phi_{fp} = V_t \ln \frac{N_a}{n_i}$	20-21 4(b)
Inversion (5)	**Threshold inversion**	$\phi_s = 2\phi_{fp}$	20-21 6(c)
	Maximum depletion width	$x_{dT} = \left[\frac{4\epsilon_s \phi_{fp}}{eN_a}\right]^{1/2}$	20-21 6(c)

Symbols:

ϕ_s = surface potential (band bending) ↩

Questions:

No.	Paper	Marks	Type	Question
Band diagrams				
1	20-21 4(b)	5	theory	Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed.
Double gate				
2	20-21 6(a)	5	theory	Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS?
Inversion				
3	20-21 6(c)	5	theory	Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram.

4.8. Bipolar transistor:

10 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 12.

Most often theory (2 of 2). Subtopics: Operating modes; Minority distribution.

Formulas:

Subtopic	Formula	Equation	Used in
Operating modes (5)	**Equilibrium minority carriers**	$p_{E0} = \frac{n_i^2}{N_E}$, $n_{B0} = \frac{n_i^2}{N_B}$, $p_{C0} = \frac{n_i^2}{N_C}$	20-21 4(c)
	Four modes	active: $V_{BE} > 0$, $V_{CB} > 0$; saturation: $V_{BE} > 0$, $V_{CB} < 0$; cutoff: $V_{BE} < 0$, $V_{CB} > 0$; inverse: $V_{BE} < 0$, $V_{CB} < 0$	20-21 4(c)
Minority distribution (5)	**Base edges**	$n_B(0) = n_{B0} \exp\left(\frac{V_{BE}}{V_t}\right)$, $n_B(x_B) = n_{B0} \exp\left(\frac{V_{BC}}{V_t}\right)$	20-21 5(b)
	Emitter and collector edges	$p_E(0) = p_{E0} \exp\left(\frac{V_{BE}}{V_t}\right)$, $p_C(0) = p_{C0} \exp\left(\frac{V_{BC}}{V_t}\right)$	20-21 5(b)

Symbols:

N_E, N_B, N_C = emitter, base, collector doping ↩ x_B = neutral base width ↩

Questions:

No.	Paper	Marks	Type	Question
Operating modes				
1	20-21 4(c)	5	theory	Show the four operating modes of a bipolar transistor as a function of V_{BE} and V_{CB}. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities.
Minority distribution				
2	20-21 5(b)	5	theory	For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector?

5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Semiconductor in equilibrium				
1	20-21 3(a)	5	Carrier concentrations	Substitute the mass action law; solve the quadratic; keep the positive root.
Carrier transport				
2	20-21 2(b)	5	Einstein relation	Set the current of a graded sample to zero; differentiate n; relate E_c to the field.
PN junction				
3	20-21 2(c)	0	Depletion width	Integrate Poisson's equation for the field and potential; use charge balance; add the edges.

6. Repeat tracker:

No repeats among these papers.

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical
Quantum basics	·	·	2
Semiconductor in equilibrium	2	1	2
Carrier transport	3	1	·
Excess carriers	2	·	1
PN junction	1	1	2
Tunnel diode	1	·	·
MOS structure	3	·	·
Bipolar transistor	2	·	·

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Semiconductor in equilibrium	30	21 %
2	Carrier transport	25	38 %
3	PN junction	25	55 %
4	Excess carriers	20	69 %
5	Quantum basics	15	79 %
6	MOS structure	15	90 %
7	Bipolar transistor	10	97 %
8	Tunnel diode	5	100 %

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